**STAC WARS | PS 2**

# Reading the Sun: Solar Radiation Analysis

Report on the Voyage-123 solar observation data.

**The short version.** Voyage-123 collected 32,686 readings, one every five minutes, over 118 days from 1 September to 31 December 2016. Solar radiation peaks around noon, from 12:00 to 13:00, and is almost zero at night. Temperature moves closely with solar radiation (correlation of 0.74), while humidity moves in the opposite direction (correlation of -0.23). The strongest solar radiation reading was 1601.26 on 4 September 2016 at 12:15, which came 6 hours 7 minutes after sunrise.

## 1. How we prepared the data

The probe logged radiation, temperature, humidity, air pressure, wind speed and direction, plus sunrise and sunset times for each day. The raw file kept the date and clock time in separate columns, so our code first joined them into a single proper date-and-time value (Pandas *to_datetime*) and did the same for sunrise and sunset. It then sorted the readings in time order and checked for problems. The data turned out to be clean: no missing dates, no missing or invalid sunrise/sunset values and no duplicate rows, so all 32,686 rows were kept. Readings arrive every five minutes (give or take a second).

## 2. Part A: Visualisation (what the data looks like)

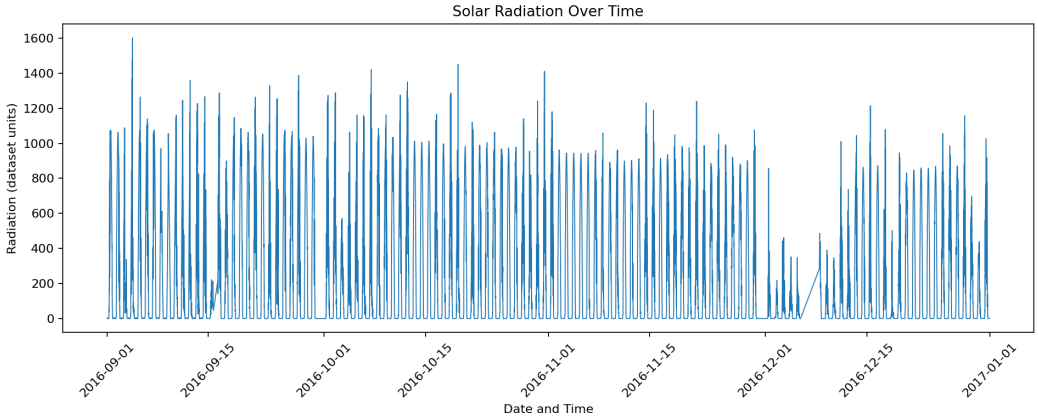



**Figure 1.** Solar radiation over the 118 days. Each spike represents one day's sunshine; the gaps between spikes are the nights.

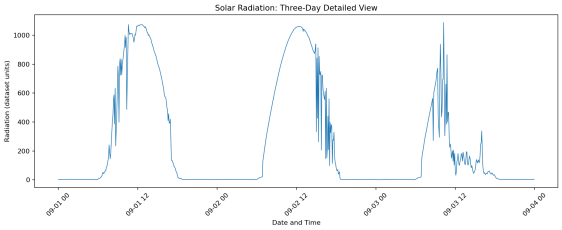



**Figure 2.** Zoom on three days: a smooth rise, a midday peak and a fall to near zero at night.

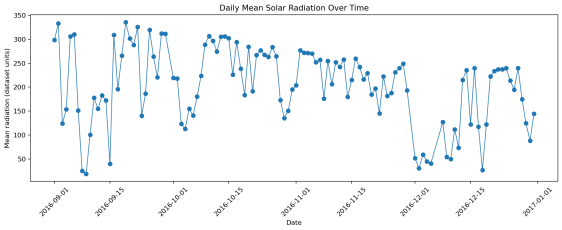



**Figure 3.** The average solar radiation of each day. It bounces around from day to day.

**What we see:** every day follows the same "hill" shape. Solar radiation is flat at night (about 1.2), climbs after sunrise, peaks near midday and drops again before sunset. Daily averages vary a lot from one day to the next.

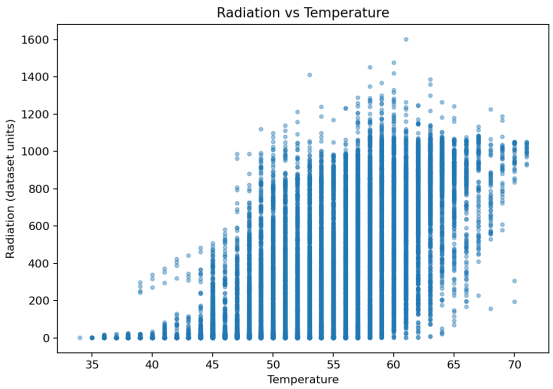



**Figure 4.** Solar radiation vs temperature: warmer readings go with sunshine.

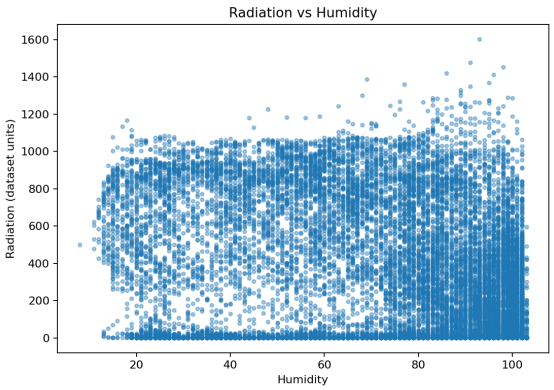



**Figure 5.** Solar radiation vs humidity: the strongest sunshine only happens when the air is not too damp.

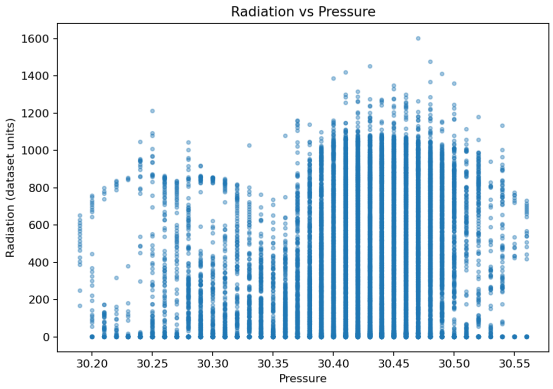



**Figure 6.** Solar radiation vs air pressure: no pattern.

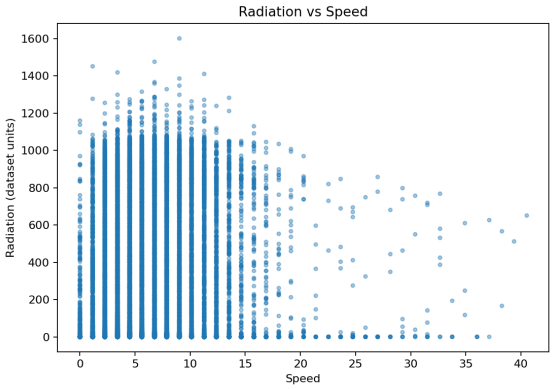



**Figure 7.** Solar radiation vs wind speed: no pattern.

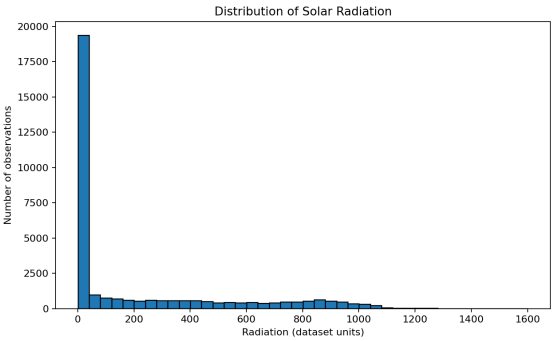



**Figure 8.** Histogram: most readings are near zero (night); fewer are high.

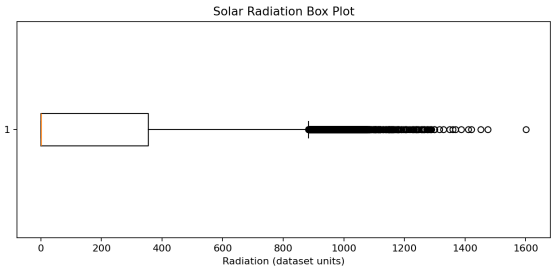



**Figure 9.** Box plot: the dots on the right are the "high" readings.

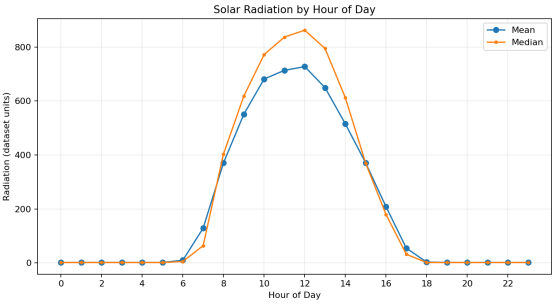



**Figure 10.** Average (mean) and typical (median) solar radiation for each hour of the day.

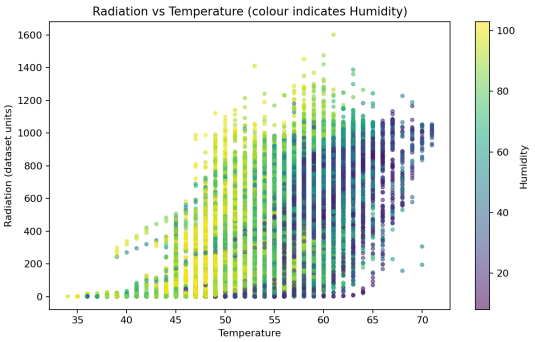



**Figure 11.** Solar radiation vs temperature, coloured by humidity (dark = damp, bright = dry).

Four environmental parameters (temperature, humidity, pressure, wind speed) are plotted against radiation, more than the two required. The scatter plots show trends over time, the histogram and box plot show the spread, and the box plot highlights the outliers.

**Bonus visualisations.** Figure 11 adds a dimension by colouring each dot by humidity, and Figures 12 and 13 compare solar radiation across humidity bands and between daylight and night:

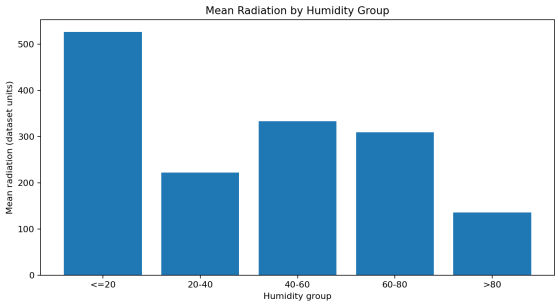



**Figure 12 (Bonus).** Average solar radiation in each humidity band. Dry air is brightest; damp air is dimmest.

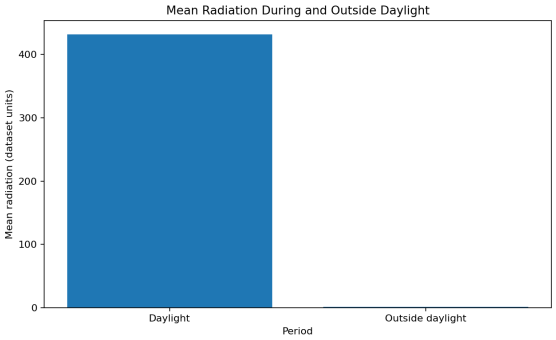



**Figure 13 (Bonus).** Average solar radiation between sunrise and sunset vs outside that window.

# 3. Part B: What the data tells us

## Q1. When is sunshine strongest?

**Answer:** Around noon, from 12:00 to 13:00, with an average of 727 and a typical (median) value of 862. The hours either side are close behind (11:00 to 12:00 averages 714 and 13:00 to 14:00 averages 650). From 07:00 onwards radiation climbs quickly, and by 18:00 it has fallen back to about 3. Between sunrise and sunset the average reading is 432; outside that window it is 1.4. Averaging the hours from 06:00 to 18:00 gives 383, versus 1.35 for the rest of the day.

## Q2. Which parameter is strongly linked to solar radiation?

A correlation runs from -1 to +1. The closer to either end, the stronger the link between two things. Temperature has the strongest correlation, so it is the clear winner. Note that this does not mean heat creates sunshine: correlation does not prove cause.

| Parameter | Correlation (r) | Meaning in words |
|---|---|---|
| Temperature | +0.735 | Strong: warmer usually means more sunshine |
| Humidity | -0.226 | Weak: damper air usually means less sunshine |
| Air pressure | +0.119 | Very weak |
| Wind speed | +0.074 | Very weak |

## Q3. How do radiation, temperature and humidity fit together?

| Humidity band | Readings | Avg radiation | Avg temperature | Temp-radiation link (r) |
|---|---|---|---|---|
| 20 or less | 632 | 526.5 | 60.6 | 0.75 |
| 20 to 40 | 4,698 | 222.1 | 52.9 | 0.87 |
| 40 to 60 | 4,007 | 333.4 | 52.8 | 0.86 |
| 60 to 80 | 5,169 | 309.3 | 51.9 | 0.77 |
| Above 80 | 18,180 | 135.3 | 49.7 | 0.52 |

The driest air (20% or less) has the most sunshine on average (526.5) and the dampest air (above 80%) the dimmest (135.3). Moisture and cloud block sunlight. Temperature and radiation stay tightly linked in every humidity band, and the link is weakest when humidity is very high (r = 0.52). Humidity and temperature are also slightly opposite each other (r = -0.29): warmer periods tend to be drier. More than half of all readings (18,180) are above 80% humidity.

## Q4. Are there unusual readings (outliers)?

We used the IQR rule, which flags readings far from the middle 50% of the data. The normal range is up to 883.7. The rule flagged 1,965 readings, all high and none low. Key points:

- **They are real sunshine, not errors.** None of the flagged readings fell outside the sunrise-to-sunset window, and the ten highest all happened around midday (11:00 to 12:40). The rule flags them mainly because over half of all readings are night-time values near zero, which pulls the "normal" range down. About 1 in 8 daytime readings (1,965 of 15,608) is flagged.

- **The top readings are suspiciously high.** The biggest readings (1601.26 on 4 Sep, 1475.40 on 4 Sep and 1451.41 on 19 Oct) happened when humidity was very high, 91% to 98%. Sunshine that strong in damp air is unusual. A likely explanation, which we cannot prove from this data, is that sunlight bounced off the edges of clouds and briefly added to direct sunlight. A sensor glitch is another possible reason.
- Across all flagged readings the average temperature was 61, the average humidity 59% and the average wind speed about 7, so these were moderately dry and breezy moments.

### Bonus insights (non-obvious findings)

- **The record-breaking day was not the brightest day overall.** 4 September has the record reading of 1601.26, but its daily average was only 153.6, far below the five brightest days (312 to 336). The record was a spike, not a long sunny day, so peaks and averages tell different stories.
- **Humidity changes how closely temperature follows the sun.** In moderately damp air the temperature-radiation link is strong (0.75 to 0.87), but in very damp air it drops to 0.52, consistent with cloud and moisture getting in the way.

## 4. Part C: Day-by-day report (Pandas and NumPy)

Our code joined the date and time columns into a datetime value, grouped all readings by date and calculated each day's maximum and average radiation for all 118 days. The full table is saved in *outputs/*.

### The record-breaking day: 4 September 2016

| Question | Answer |
|---|---|
| Highest radiation recorded | 1601.26 |
| Time of the peak | 12:15 (12:15:04) |
| Temperature / humidity / wind speed at the peak | 61 / 93% / 9.0 |
| Sunrise and sunset that day | 06:08 and 18:35 (day length 12 h 27 min) |
| Peak came this long after sunrise | 6 hours 7 minutes |
| ...and this long before sunset | 6 hours 19 minutes |

So the peak came very close to the middle of the day (the halfway point between sunrise and sunset is about 6 h 14 min). The top five days by average solar radiation are listed below. These are not the record-peak days; they are simply the days that were bright for the longest periods.

| Rank | Date | Avg radiation | Highest reading that day |
|---|---|---|---|
| 1 | 19 Sep 2016 | 335.5 | 1085.75 |
| 2 | 2 Sep 2016 | 333.5 | 1062.01 |
| 3 | 22 Sep 2016 | 325.5 | 1053.17 |
| 4 | 25 Sep 2016 | 319.9 | 1075.29 |
| 5 | 28 Sep 2016 | 312.4 | 1029.30 |

## 5. Conclusion

The sun on Voyage-123's world follows a daily rhythm. It is dark all night, then climbs steadily after sunrise, reaches a peak around noon and falls by evening. Temperature is the parameter most closely tied to sunshine. Humidity acts as a brake: the dampest air gives the least sunshine. The record reading of 1601.26 on 4 September looks like a spike on an otherwise ordinary day. For the Rebellion's planners: expect peak power from about eleven o'clock to two o'clock, expect low output on damp days, and treat sudden spikes with care.

**Files.** All results above were produced by *solar_analysis.py*, which uses Pandas, NumPy and Matplotlib. Run it in the folder that contains *SolarPrediction.csv*. It creates a *figures/* folder with all plots and an *outputs/* folder with all tables.

**Limitation:** the data covers four months, from September to December 2016, so we cannot say anything about other seasons.

May the code be with you!